

App Dev Project Report

1. Student Details

Name: Akash Maurya

Roll Number: 23f1001713

Email: 23f1001713@ds.study.iitm.ac.in

About Me: I am a diploma student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

Placement Portal Application web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows students and companies to register and then login using flask-login for authentication.

3. AI/LLM Declaration

I used **claude code** to assist in writing some css styles for the student dashboard.

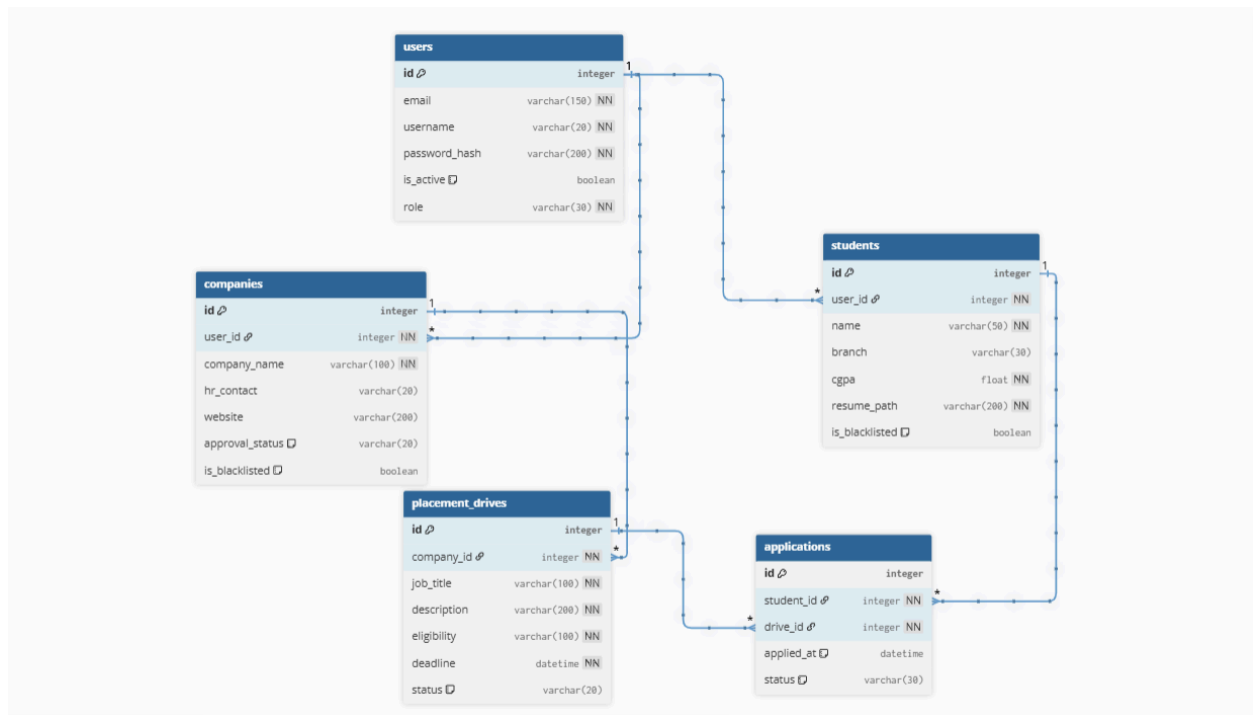
The extent of AI/LLM usage is around **less than 10%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data
datetime	For application or drive deadline in database
Werkzeug	For password hashing

5. Database Schema / ER Diagram



Relationships:

Users → Companies

One **user** can be linked to **one company** (via `companies.user_id`).

Users → Students

One **user** can be linked to **one student** (via `students.user_id`).

Companies → Placement Drives

- A single **company** can organize multiple **placement_drives**.
- Each drive belongs to exactly one company.
- So: **One company → Many placement drives**

Placement Drives → Applications

- A single **placement_drive** can have many **applications**.
- Each application belongs to exactly one drive.
- So: **One placement drive → Many applications**

Students → Applications

- A single **student** can apply to multiple drives.
- Each application belongs to exactly one student.
- So: **One student → Many applications**

7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point and has all the routes
- **/models** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates

Implemented Features:

User registration and login
Role based dashboard access and login protected routes
Secure password hashing and user session handling

Search feature for company and student

8. Video Presentation

Drive Link:

 VN20260228_140843.mp4